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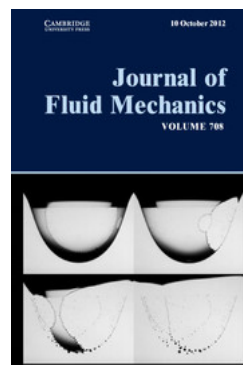
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A Stokesian analysis of a submerged viscous jet impinging on a planar wall

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The wall pressure and wall shear stress of a submerged viscous jet impinging on an infinite planar wall are derived. The whole creeping flow of semi-infinite extent is generated via distributions on a cylindrical pipe of tangentially and normally directed Stokeslets which are modified to achieve no-slip at the wall in two stages. First the pressure and vorticity jumps associated with the Poiseuille flow upstream in the pipe are readily forced, and then further distributions, of zero density far upstream but with square-root density singularity at the orifice $z = h$, are added to achieve no-slip on the pipe wall. Thus the adjustment of the interior pipe flow from its upstream parabolic profile to its exit profile is fully included in – and a major feature of – this creeping flow analysis. The maximum plane wall pressure is always located on the axis $r = 0$, and decreases as h increases to alleviate the obstruction effect of the wall. The interaction of the inflow with the ambient fluid in the neighbourhood of $z = 0$ causes the wall stress to rise rapidly to a maximum and then decay with the radial position of this maximum increasing as h increases. This behaviour is discussed in the context of physiological experiments on auditory sensory hair cells that motivated this study.

Key words: low-Reynolds-number flows, Stokesian dynamics

1. Introduction

Physiological experiments often involve the application of a submerged viscous fluid jet impinging on biological cells (Deshpande & Vaishnav 1982, 1983; Richards *et al.* 1995; Kong, Bursac & Tung 2005; Bouafsoun *et al.* 2006). In particular, biophysical, i.e. electrical, chemical and mechanical, properties of auditory sensory hair cells have been measured while perfusing them with extracellular solution via a pipette completely submerged in a water-based salt solution which bathes the cell and maintains its integrity. In most cases, the pipette was aimed at the lateral wall of the cell. Three different experiments have indicated that the magnitude of

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the physicochemical response to the perfusate, i.e. extracellular solution, correlates with the distance from the orifice of the pipette to the cell. Firstly, different groups have reported different magnitudes of the whole-cell ionic current induced by extracellular solution containing adenosine triphosphate (Nakagawa *et al.* 1990; Housley, Greenwood & Ashmore 1992; Lin, Hume & Nuttall 1993), but closer examination suggests that larger currents were obtained if the pipette was placed close to the cell. Secondly, the magnitude of the cellular nonlinear electrical capacitance (charge movement across the cell wall) in response to extracellular solution containing chloride increased if the pipette was placed close to the cell (Santos-Sacchi *et al.* 2006). Thirdly, increased hydraulic conductivity, or equivalently rate of volume change due to water intake accompanied by a reduction in the ratio of the principal values of the anisotropic i.e. orthotropic strains of the cell wall, was observed when hypo-osmotic extracellular solution was applied close to the cell (Morimoto *et al.* 2002; Zhi *et al.* 2007; Farahbakhsh, Zelaya & Narins 2010). Reviews (Brownell 2006; Spector *et al.* 2006) hypothesize that the increase in the magnitude of the wall pressure and wall shear stress along the lateral wall of the cell due to the reduced distance of the wall from the orifice alters the biophysical properties of the wall, resulting in the increase in cellular response to the perfusate. Since alterations in the biophysical properties of the hair cell can result in diminished selectivity and sensitivity of hearing, analysis of the wall pressure and wall shear stress induced by a pipette could contribute to the understanding of the cellular mechanisms of sensorineural hearing loss, i.e. deafness (Ashmore *et al.* 2010).

In these experiments, the tapered end of the pipette was 3–5 mm long with an orifice of diameter 5–50 μm at a distance of 10–100 μm directly above the cell. The small aspect ratio of the tapered end suggests that the flow out of the orifice is both fully developed and slow, with a velocity of $\sim 2\text{--}5\text{ mm s}^{-1}$ (Saunders & Syzmko 1989; Dinklo, Meulenberg & van Netten 2007). The slow flow ensures that the cell does not disappear out of view under the microscope, and is completely perfused. Since the bathing solution is typically a water-based salt solution, the resulting Reynolds number is of the order of 10^{-3} . Thus, as a first approximation, it is necessary to determine how the wall pressure and wall shear stress for an axisymmetric Stokesian fluid jet impinging on a planar wall behaves as a function of the distance of the orifice from the wall.

Here the flow is assumed to be purely viscous, in contrast to models of non-zero Reynolds number flow, which use methods that depend on the presence of inertia and usually assume a parabolic or uniform exit flow (Akaike & Nemoto 1998; Baydar 1999; Bernard, Brizzi & Bousgarbies 2000; Phares, Smedley & Flagan 2000*a,b*). In particular, Stokeslet singularities (Happel & Brenner 1983; Davis 1991) are used here in a Stokesian model of the impinging submerged viscous fluid jet that includes full consideration of flow within the pipe. Figure 1 illustrates the axisymmetric impinging jet on a planar wall caused by upstream Poiseuille (i.e. parabolic) flow that, after significant changes to its structure, emerges from a cylindrical pipe into the same viscous fluid of semi-infinite extent. All spatial dimensions have been normalized by the radius of the pipe, whose orifice is held at a distance h radii above the wall. The goal is to obtain, with full account taken of the flow within the pipe, expressions for the stress and pressure at the plane wall $z = 0$ for $r \geq 0$, and examine their form for different aspect ratios $h/2$ of order unity without recourse to artificial boundaries at infinity.

Creeping flows involving a semi-infinite or finite pipe geometry have usually been determined by use of the superposition method (Meleshko 1996) or in terms

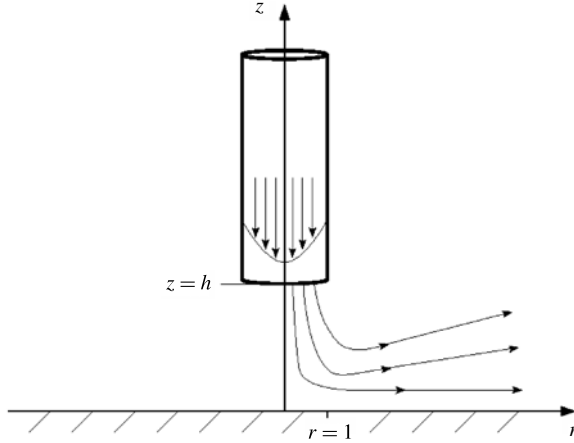


FIGURE 1. Schematic of the mathematical problem. The pipe wall ($r = 1$, $z > h$) is assumed to be infinitesimally thin. Parabolic flow is indicated upstream in the interior of the semi-infinite pipe.

of Papkovitch–Fadle pipe eigenfunctions (Joseph & Sturges 1978). Each approach poses significant difficulties in calculating the unknown coefficients. The former has a non-uniqueness disadvantage (Davis 2002) while the latter lacks orthogonality (Spence 1983) and cannot exploit the ever more rapid pipewise decay of the set of eigenfunctions. For the computation of streamlines a ‘least-squares’ procedure for achieving no-slip has advantages, particularly for some non-standard domains (Shankar 2005). When additional rigid boundaries are present, as here, the difficulties associated with matching series of functions over a common bounding surface are multiplied (Trogdon & Joseph 1982). Here a different solution construction is achieved by focusing attention on the stress discontinuities along the pipe wall. Axisymmetric flows are generated by rings of Stokeslets directed along or normal to the wall (Davis 2009). Since the upstream flow creates the viscous jet, the first, relatively straightforward, step is to establish the ring singularities that yield the prescribed Poiseuille flow. After modifying these to achieve no-slip at the rigid plane, $z = 0$, the resulting reflected velocity field must be annihilated at both the inside and outside of the rigid pipe wall by further use of ring singularities, this being the second, more difficult, step in the analysis. Its purpose is to determine the adjustment of the pipe flow from its upstream parabolic profile to its exit profile, a procedure that is equivalent to including the Papkovitch–Fadle pipe eigenfunctions. This classic difficulty in creeping flow is absent at high Reynolds number, which allows the uniform or parabolic exit profile approximation and thereby circumvents consideration of flow within the pipe. The additional ring singularities have distribution densities that evidently vanish far upstream, and are anticipated to have inverse square-root behaviour at the orifice. Guided by earlier work (Davis 1991, 2003), Abel transforms are introduced and the no-slip conditions are manipulated into a pair of integral equations of Fredholm type, whose solution yields, in principle but indirectly, the density functions. The kernels of the integral equations are themselves integrals involving a product of a modified Bessel function and a pair of Bessel functions of the first or second kind. The necessary computation needed not only the conversion of the Lucas algorithm (Lucas 1995) to MATLAB but also its extension to Bessel

functions of the second kind (Ratnanather *et al.* 2012). This careful consideration of the stress singularity at the orifice rim does not feature in standard computational fluid dynamics codes.

The rest of the paper is outlined as follows. Section 2 describes the mathematical model. Section 3 describes the numerical method used to solve the mathematical model. Section 4 describes the results. The paper concludes with a brief discussion in § 5 and adds an Appendix that amplifies some analytical details of the integral equations.

2. Mathematical model

2.1. Preliminaries

The approach is to generate the whole flow by using distributions on the cylinder ($r = 1, z > h$) of tangentially and normally directed Stokeslets which are readily modified to achieve no-slip at the plane wall $z = 0$. Two principal stages are required, and their development may be regarded as successive modifications of the stream function to more closely represent the whole flow. Since the upstream forcing is the jet's driving mechanism, it must be the fundamental contributor, with prescribed scaling for uniqueness. So first the pressure and vorticity jumps associated with the Poiseuille flow upstream in the pipe $r < 1, z > h$ are imposed and modified to take account of the plane wall at $z = 0$. Then further distributions, of zero density far upstream but with square-root density singularity at $z = h$, are added to achieve no-slip on the pipe wall ($r = 1$).

Thus consider creeping flow bounded by the wall $z = 0$ and generated by pressure-driven flow in the semi-infinite fixed cylinder $r = 1, z > h$. The velocity and dynamic pressure fields are governed by the Stokes equations

$$\mu \nabla^2 \mathbf{v} = \nabla p, \quad \nabla \cdot \mathbf{v} = 0, \quad (2.1)$$

where μ is the coefficient of viscosity. Axisymmetric flow allows the velocity representation, with obvious unit vectors,

$$\mathbf{v} = \frac{1}{r} \frac{\partial \psi}{\partial r} \hat{\mathbf{z}} - \frac{1}{r} \frac{\partial \psi}{\partial z} \hat{\mathbf{r}} \quad (2.2)$$

in terms of the stream function ψ that satisfies

$$L_{-1}^2 \psi = \left(\frac{\partial^2}{\partial r^2} - \frac{1}{r} \frac{\partial}{\partial r} + \frac{\partial^2}{\partial z^2} \right)^2 \psi = 0. \quad (2.3)$$

The pressure p and the vorticity $-r^{-1} L_{-1} \psi$ are then related by

$$\frac{\partial p}{\partial z} = \frac{\mu}{r} \frac{\partial}{\partial r} L_{-1} \psi, \quad \frac{\partial p}{\partial r} = -\frac{\mu}{r} \frac{\partial}{\partial z} L_{-1} \psi. \quad (2.4)$$

No-slip and the imposed pipe outlet flow force ψ to satisfy the boundary conditions

$$\psi = -1, \quad \frac{\partial \psi}{\partial r} = 0 \quad \text{at } r = 1, z > h, \quad (2.5)$$

$$\psi = 0 = \frac{\partial \psi}{\partial z} \quad \text{at } z = 0, \quad (2.6)$$

$$\psi = O(r^2) \quad \text{as } r \rightarrow 0. \quad (2.7)$$

Conditions (2.5) and (2.7) ensure that

$$\psi(r, z) \sim \psi_\infty(r) = -2r^2 + r^4 \quad \text{as } z \rightarrow \infty \quad (0 \leq r \leq 1), \quad (2.8)$$

which is the far upstream Poiseuille flow that creates the ‘jet-like’ motion through the orifice at $z = h$, $0 \leq r < 1$. This scaling of the far upstream flow to have dimensionless flux 2π and mean velocity 2 is relevant to the dimensionless wall pressure and wall stress displayed in §4. The proximity h of the orifice to the plane wall is assumed to be of the same order as the pipe diameter, 2, since only such values are of interest in the physiological application considered here. The chosen mathematical method has the same range of applicability and may be regarded as bridging the gap between asymptotic results for large and small h .

2.2. Upstream forcing

Following Davis (1996, 2009), Stokeslet singularities are distributed along the cylinder wall and modified to accommodate the no-slip condition (2.6). Davis (1996) noted that known flow fields around disks can be identified as due to normally and radially distributed Stokeslets, because it is readily shown that a ring of force singularities of strength $4\mu\hat{z}$ (sign error corrected) at $z = 0$, $r = r'$ generates the stream function

$$\psi_{ax}(r, r'; z) = \int_0^\infty r J_1(kr) J_0(kr') e^{-k|z|} (k^{-1} + |z|) dk, \quad (2.9)$$

while a ring of force singularities of strength $-4\mu\hat{r}$ at $z = 0$, $r = r'$ generates the stream function

$$\psi_{rad}(r, r'; z) = \int_0^\infty r J_1(kr) J_1(kr') e^{-k|z|} z dk. \quad (2.10)$$

Alternative forms of these integrals, required here for the identification of discontinuities across the cylinder wall, are obtained from the identity

$$\int_0^\infty J_0(kr) e^{-k|z|} dk = (r^2 + z^2)^{-1/2} = \frac{2}{\pi} \int_0^\infty K_0(kr) \cos kz dk \quad (2.11)$$

and the Fourier series of $J_0[k(r^2 + r'^2 - 2rr' \cos \phi)^{1/2}]$ and $K_0[k(r^2 + r'^2 - 2rr' \cos \phi)^{1/2}]$. They are

$$\psi_{ax} = \frac{2}{\pi} \int_0^\infty r [r K_0(kr_>) I_0(kr_<) - r' K_1(kr_>) I_1(kr_<)] \cos kz dk \quad (2.12)$$

and

$$\psi_{rad} = \frac{2}{\pi} \int_0^\infty r K_1(kr_>) I_1(kr_<) z \cos kz dk = -\frac{2}{\pi} \int_0^\infty r \frac{\partial}{\partial k} [K_1(kr_>) I_1(kr_<)] \sin kz dk, \quad (2.13)$$

where $r_> = \max(r, r')$, $r_< = \min(r, r')$. Thus there are two types of ring singularities, denoted by subscripts (*ax*, *rad*), which must be suitably combined here.

From (2.12) and (2.13), it is seen that p_{ax} and $L_{-1}\psi_{rad}$ are continuous at $r = r'$ but

$$[L_{-1}\psi_{ax}]_{r=r'}^{r'+} = -4\delta(z) = \mu^{-1}r'[p_{rad}]_{r=r'}^{r'+}. \quad (2.14)$$

In the Poiseuille flow (2.8), $L_{-1}\psi_\infty = 8r^2$, $p_\infty = 16\mu z + \text{constant}$, and so the required discontinuities can be achieved by constructing

$$\begin{aligned}\psi_1(r, z) &= 2 \int_h^\infty \psi_{ax}(r, 1; z - z_0) dz_0 + 4 \int_h^\infty (z_0 - h) \psi_{rad}(r, 1; z - z_0) dz_0 \\ &= -2r \int_0^\infty J_1(kr) J_2(k) \\ &\quad \times \left\{ \frac{4}{k^2} H(z - h) - \left[\frac{2}{k^2} \text{sgn}(z - h) + \frac{z - h}{k} \right] e^{-k|z - h|} \right\} dk, \quad (2.15)\end{aligned}$$

where $H(x)$ denotes the Heaviside unit function. The evaluation of the corresponding pressure field $p_1(r, z)$ is achieved by reverting to the forms (2.9) and (2.10), which yield, by use of (2.4),

$$\begin{aligned}p_1(r, z) &= 2 \int_h^\infty [p_{ax}(r, 1; z - z_0) + 2(z_0 - h)p_{rad}(r, 1; z - z_0)] dz_0 \\ &= 4\mu \int_0^\infty k J_0(kr) \int_h^\infty e^{-k|z - z_0|} [\text{sgn}(z - z_0) J_0(k) + 2(z_0 - h) J_1(k)] dz_0 dk \\ &= 4\mu \int_0^\infty J_0(kr) [J_2(k) e^{-k|z - h|} + 2(z - h + |z - h|) J_1(k)] dk \\ &= 4\mu \int_0^\infty J_0(kr) J_2(k) e^{-k|z - h|} dk + 8\mu(z - h + |z - h|) H(1 - r). \quad (2.16)\end{aligned}$$

Hence $p_1 \sim 16\mu(z - h)H(1 - r)$ as $z \rightarrow \infty$, as required to ensure $\psi_1(r, z) \sim \psi_\infty(r)H(1 - r)$, the far upstream flow. For ψ_1 to yield no-slip along the entire pipe wall, the lower limit h in (2.15) would need to be replaced by $-\infty$. Values of ψ_1 at either end of the pipe are given, from (2.15), by

$$\psi_1(r, \infty) = 2\psi(r, h) = \begin{cases} r^4 - 2r^2 & (r < 1), \\ -1 & (r > 1). \end{cases} \quad (2.17)$$

Thus only half of the upstream pipe flux generated by ψ_1 reaches the pipe exit, the rest seeping through the pipe wall, particularly near the orifice. Though the axial flow is parabolic at $z = h$, the radial velocity is $rF(3/2, -1/2, 2, r^2)$, where F is a hypergeometric function. Thus a semi-infinite pipe, as here, needs an ‘exit flow’ adjustment to the stream function, whether or not the plane wall is present.

The no-slip condition (2.6) at the plane wall is evidently satisfied by the stream functions

$$\begin{aligned}\Psi_{rad}(r, z; z_0) &= \psi_{rad}(r, 1; z - z_0) + \int_0^\infty r J_1(kr) J_1(k) e^{-k(z+z_0)} (z_0 - z + 2kz z_0) dk, \\ &= \psi_{rad}(r, 1; z - z_0) + \psi_{rad}(r, 1; z + z_0) \\ &\quad - 2z \int_0^\infty r J_1(kr) J_1(k) e^{-k(z+z_0)} (1 - kz z_0) dk, \quad (2.18)\end{aligned}$$

$$\begin{aligned}\Psi_{ax}(r, z; z_0) &= \psi_{ax}(r, 1; z - z_0) - \int_0^\infty r J_1(kr) J_0(k) e^{-k(z+z_0)} (k^{-1} + z_0 + z + 2kz z_0) dk, \\ &= \psi_{ax}(r, 1; z - z_0) - \psi_{ax}(r, 1; z + z_0) \\ &\quad - 2z z_0 \int_0^\infty r J_1(kr) J_0(k) e^{-k(z+z_0)} k dk, \quad (2.19)\end{aligned}$$

which are modifications of (2.9) and (2.10). Hence, following (2.15), the upstream Poiseuille flow stream function is modified to

$$\begin{aligned}\Psi_1(r, z) &= 2 \int_h^\infty \Psi_{ax}(r, z; z_0) dz_0 + 4 \int_h^\infty (z_0 - h) \Psi_{rad}(r, z; z_0) dz_0 \\ &= 2 \int_0^\infty r J_1(kr) \frac{J_2(k)}{k^2} \{-2 \operatorname{sgn}(z - h) [1 - e^{-k|z-h|}] + k(z - h) e^{-k|z-h|} \\ &\quad - 2[1 - e^{-k(z+h)}] + k(3z + h + 2khz) e^{-k(z+h)}\} dk. \quad (2.20)\end{aligned}$$

This arrangement displays three contributions arising respectively from the prescribed ring singularities, their images directed to achieve zero normal flow at $z = 0$ and the integrals in (2.18) and (2.19) that serve to also achieve zero tangential flow at $z = 0$. It is this combination of images that produces the common $J_2(k)$ factor in (2.20), as seen by using the definitions (2.9) and (2.10) to write

$$\begin{aligned}2 \int_h^\infty \psi_{ax}(r, 1; z + z_0) dz_0 - 4 \int_h^\infty (z_0 - h) \psi_{rad}(r, 1; z + z_0) dz_0 \\ = 2 \int_h^\infty \psi_{ax}(r, 1; -z - z_0) dz_0 + 4 \int_h^\infty (z_0 - h) \psi_{rad}(r, 1; -z - z_0) dz_0. \quad (2.21)\end{aligned}$$

Note that Ψ_1 may be verified to be continuous at $z = h$ up to the third derivative, as expected of a solution of the fourth-order equation (2.3). Moreover, $\Psi_1 \sim [-1 + (1 - r^2)^2 H(1 - r)]$ as $z \rightarrow \infty$, giving, for $r < 1$, the required limit $\psi_\infty(r)$ in (2.8) for the far upstream Poiseuille flow. Also, within the fluid,

$$\begin{aligned}L_{-1}\Psi_1(r, z) &= 4r^2 H(1 - r) [1 + \operatorname{sgn}(z - h)] \\ &\quad - 4 \int_0^\infty r J_1(kr) J_2(k) [e^{-k|z-h|} \operatorname{sgn}(z - h) + e^{-k(z+h)} (3 + 2kh)] dk, \quad (2.22)\end{aligned}$$

$$\begin{aligned}P_1(r, z) &= 8\mu H(1 - r) [z - h + |z - h|] \\ &\quad + 4\mu \int_0^\infty J_0(kr) J_2(k) [e^{-k|z-h|} + e^{-k(z+h)} (3 + 2kh)] dk. \quad (2.23)\end{aligned}$$

This completes the first stage of the flow-field construction, making available a stream function Ψ_1 with $\Psi_1 \sim \psi_\infty(r) H(1 - r)$ as $z \rightarrow \infty$ and zero velocity at $z = 0$.

The far-field forms of the above stream functions can be evaluated for $r \gg 1$ by setting $J_0(k) \sim 1$, $J_1(k) \sim k/2$, $J_2(k) \sim k^2/8$ in the relevant integrals. Thus (2.9) and (2.10) yield

$$\psi_{rad}(r, 1; z) \sim \frac{r^2 z}{2(r^2 + z^2)^{3/2}}, \quad \psi_{ax}(r, 1; z) \sim \frac{r^2}{(r^2 + z^2)^{1/2}} \quad (r \gg 1), \quad (2.24)$$

and hence (2.18) and (2.19) give

$$\begin{aligned}\Psi_{rad}(r, z; z_0) &\sim \frac{r^2(z - z_0)}{2[r^2 + (z - z_0)^2]^{3/2}} + \frac{r^2(z + z_0)}{2[r^2 + (z + z_0)^2]^{3/2}} \\ &\quad - \frac{r^2 z}{[r^2 + (z + z_0)^2]^{3/2}} \left[1 - \frac{3z_0(z + z_0)}{r^2 + (z + z_0)^2} \right] \quad (r \gg 1), \quad (2.25)\end{aligned}$$

$$\Psi_{ax}(r, z; z_0) \sim \frac{r^2}{[r^2 + (z - z_0)^2]^{1/2}} - \frac{r^2}{[r^2 + (z + z_0)^2]^{1/2}} - \frac{2r^2 z z_0}{[r^2 + (z + z_0)^2]^{3/2}}, \quad (2.26)$$

which respectively display a Stokes doublet and a Stokeslet at $(0, z_0)$, together with their images in the plane wall. Similarly, (2.20) gives

$$\begin{aligned} \Psi_1(r, z) \sim & -\frac{z-h}{2[r^2+(z-h)^2]^{1/2}} - \frac{z+h}{2[r^2+(z+h)^2]^{1/2}} + \frac{r^2(z-h)}{4[r^2+(z-h)^2]^{3/2}} \\ & + \frac{r^2(3z+h)}{4[r^2+(z+h)^2]^{3/2}} + \frac{3r^2zh(z+h)}{2[r^2+(z+h)^2]^{5/2}} \quad (r \gg 1), \end{aligned} \quad (2.27)$$

which displays a volume source and a Stokes doublet at $(0, h)$, with their images in $z=0$. This stream function is evidently constructed as a superposition of distributions, defined by (2.15), of the two stream functions in (2.26). These estimates are valid far from the symmetry axis, with our stated restriction $h/2 = O(1)$.

2.3. No-slip forcing

In the second stage, further distributions of the above Stokeslets are used to cancel the velocity components generated by Ψ_1 at the cylinder wall $r=1, z>h$, and illustrated in figure 4. According to (2.20), these are

$$\begin{aligned} U_1(1, z) = 2 \int_0^\infty J_1(k) \frac{J_2(k)}{k} \left\{ [1+k(z-h)]e^{-k(z-h)} \right. \\ \left. + [-1+3kz+kh(2kz-1)]e^{-k(z+h)} \right\} dk \quad (z > h), \end{aligned} \quad (2.28)$$

$$\begin{aligned} W_1(1, z) = 2 \int_0^\infty J_0(k) \frac{J_2(k)}{k} \left\{ [2+k(z-h)]e^{-k(z-h)} \right. \\ \left. + [2+3kz+kh(2kz+1)]e^{-k(z+h)} \right\} dk \quad (z > h), \end{aligned} \quad (2.29)$$

which are such that $U_1(1, z) \sim 3/z^3$, $W_1(1, z) \sim 3/z^2$ as $z \rightarrow \infty$.

The additional stream function has the form

$$\Psi_2(r, z) = \int_h^\infty [C(z_0)\Psi_{rad}(r, z; z_0) + D(z_0)\Psi_{ax}(r, z; z_0)] dz_0, \quad (2.30)$$

with the density functions $C(z_0)$ and $D(z_0)$ expected to vanish at infinity, and have square-root singularities at the orifice $z=h$. Like Ψ_1 in (2.20), this representation in terms of Ψ_{rad} and Ψ_{ax} , given by (2.18) and (2.19), rather than ψ_{rad} and ψ_{ax} , ensures that Ψ_2 also yields no-slip at the plane wall $z=0$. According to (2.14), the vorticity and pressure discontinuities across the wall are given by

$$[L_{-1}\Psi_2]_{r=1-}^{1+} = -4D(z), \quad [P_2]_{r=1-}^{1+} = -4\mu C(z) \quad (z > h). \quad (2.31)$$

As with flow through a hole in a wall (Davis 1991, 2003), their edge behaviour is anticipated by setting

$$C(z) = \frac{d}{dz} \int_h^z \frac{t^{1/2}S(t) dt}{\sqrt{z^2-t^2}}, \quad D(z) = \frac{d}{dz} \int_h^z \frac{t^{1/2}T(t) dt}{\sqrt{z^2-t^2}}, \quad (2.32)$$

whose substitution in (2.30) yields, after some manipulation,

$$\Psi_2(r, z) = - \int_h^\infty t^{1/2}S(t) \int_t^\infty \frac{\partial \Psi_{rad}}{\partial z_0}(r, z; z_0) \frac{dz_0}{\sqrt{z_0^2-t^2}} dt$$

$$- \int_h^\infty t^{1/2} T(t) \int_t^\infty \frac{\partial \Psi_{ax}}{\partial z_0}(r, z; z_0) \frac{dz_0}{\sqrt{z_0^2 - t^2}} dt \quad (2.33)$$

where S, T are Abel transforms, having no obvious physical interpretation. Their important role is as bounded, continuous mappings of the density functions, C, D . The factor $t^{1/2}$ is inserted to achieve symmetric kernels in the subsequent integral equations.

A disadvantage of expressing (2.30) in the same form as (2.20) is that a possible solution is $C(z) = -4(z - h), D(z) = -2$. Indeed, since the amplitude of the pipe flow is prescribed, any non-zero multiple of this pair within a solution must be disallowed. The inversion formula shows that the corresponding Abel transforms are

$$S(t) = \frac{2t^{1/2}}{\pi} \int_h^t \frac{-4(z - h) dz}{\sqrt{t^2 - z^2}} = -\frac{8t^{1/2}}{\pi} [\sqrt{t^2 - h^2} - h \arccos(h/t)], \quad (2.34)$$

$$T(t) = -\frac{4t^{1/2}}{\pi} \arccos(h/t), \quad (2.35)$$

which are $O(t^{3/2}), O(t^{1/2})$ respectively as $t \rightarrow \infty$ and must be suppressed.

The z_0 -integrals in (2.33) will be evaluated by use of the formulas

$$\int_t^\infty \left[\begin{array}{c} \frac{2}{\pi} \sin kz \\ \frac{2}{\pi} \cos kz \\ e^{-kz} \end{array} \right] \frac{dz}{\sqrt{z^2 - t^2}} = \left[\begin{array}{c} J_0(kt) \\ -Y_0(kt) \\ K_0(kt) \end{array} \right], \quad (2.36)$$

(Gradshteyn & Ryzhik 2007), but there is a difficulty because the form $e^{-k|z-z_0|}$, arising from (2.9) and (2.10), is unsuitable for direct application of (2.36). The forms $\cos k(z - z_0)$ and $\sin k(z - z_0)$, arising from (2.12) and (2.13), are simplified here by including the image terms in (2.18) and (2.19) when switching to the alternative forms. Thus, write

$$\begin{aligned} \Psi_{rad}(r, z; z_0) &= -\frac{4}{\pi} \int_0^\infty r \frac{\partial}{\partial k} [K_1(kr_>) I_1(kr_<)]_{r'=1} \sin kz \cos kz_0 dk \\ &\quad - 2z \int_0^\infty r J_1(kr) J_1(k) e^{-k(z+z_0)} (1 - kz_0) dk, \end{aligned} \quad (2.37)$$

$$\begin{aligned} \Psi_{ax}(r, z; z_0) &= \frac{4}{\pi} \int_0^\infty r [r K_0(kr_>) I_0(kr_<) - r' K_1(kr_>) I_1(kr_<)]_{r'=1} \sin kz \sin kz_0 dk \\ &\quad - 2zz_0 \int_0^\infty r J_1(kr) J_0(k) e^{-k(z+z_0)} k dk. \end{aligned} \quad (2.38)$$

Hence, in (2.33),

$$\begin{aligned} \int_t^\infty \frac{\partial \Psi_{rad}}{\partial z_0}(r, z; z_0) \frac{dz_0}{\sqrt{z_0^2 - t^2}} &= 2 \int_0^\infty r \frac{\partial}{\partial k} [K_1(kr_>) I_1(kr_<)]_{r'=1} k J_0(kt) \sin kz dk \\ &\quad + 2z \int_0^\infty r J_1(kr) J_1(k) k [2K_0(kt) - kt K_1(kt)] e^{-kz} dk, \end{aligned} \quad (2.39)$$

$$\begin{aligned} \int_t^\infty \frac{\partial \Psi_{ax}}{\partial z_0}(r, z; z_0) \frac{dz_0}{\sqrt{z_0^2 - t^2}} &= -2 \int_0^\infty r [r K_0(kr_>) I_0(kr_<) \\ &\quad - K_1(kr_>) I_1(kr_<)]_{r'=1} k Y_0(kt) \sin kz dk \end{aligned}$$

$$- 2z \int_0^\infty r J_1(kr) J_0(k) k [K_0(kt) - kt K_1(kt)] e^{-kz} dk. \quad (2.40)$$

When (2.39) and (2.40) are substituted into (2.33), it follows that the velocity components generated by $\Psi_2(r, z)$ at the cylinder wall are given, with Bessel functions evaluated at k unless otherwise stated, by

$$\begin{aligned} U_2(1, z) = & 2 \int_h^\infty t^{1/2} S(t) \left\{ \int_0^\infty \frac{d}{dk} [K_1 I_1] k^2 J_0(kt) \cos kz dk \right. \\ & + \int_0^\infty (1 - kz) J_1^2 k [2K_0(kt) - kt K_1(kt)] e^{-kz} dk \left. \right\} dt \\ & - 2 \int_h^\infty t^{1/2} T(t) \left\{ \int_0^\infty [K_0 I_0 - K_1 I_1] k^2 Y_0(kt) \cos kz dk \right. \\ & + \int_0^\infty (1 - kz) J_1 J_0 k [K_0(kt) - kt K_1(kt)] e^{-kz} dk \left. \right\} dt \quad (z > h) \end{aligned} \quad (2.41)$$

$$\begin{aligned} W_2(1, z) = & 2 \int_h^\infty t^{1/2} S(t) \left\{ \int_0^\infty [K_0 I_0 - K_1 I_1] k^2 J_0(kt) \sin kz dk \right. \\ & - z \int_0^\infty J_0 J_1 k^2 [2K_0(kt) - kt K_1(kt)] e^{-kz} dk \left. \right\} dt \\ & + 2 \int_h^\infty t^{1/2} T(t) \left\{ \int_0^\infty \frac{d}{dk} (k^2 K_0 I_0) Y_0(kt) \sin kz dk \right. \\ & + z \int_0^\infty J_0^2 k^2 [K_0(kt) - kt K_1(kt)] e^{-kz} dk \left. \right\} dt \quad (z > h). \end{aligned} \quad (2.42)$$

2.4. Problem statement

With no-slip on the plane wall guaranteed by Ψ_1, Ψ_2 , the no-slip conditions (2.5) are now applied in the form

$$\frac{2}{\pi} \frac{d}{ds} \int_s^\infty \left[\frac{U_1(1, z) + U_2(1, z)}{W_1(1, z) + W_2(1, z)} \right] \frac{z dz}{\sqrt{z^2 - s^2}} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (s > h), \quad (2.43)$$

by use of the formulas, readily deduced by two differentiations of (2.36),

$$\frac{2}{\pi} \frac{d}{ds} \int_s^\infty \frac{z \cos kz dz}{\sqrt{z^2 - s^2}} = -ks J_0(ks), \quad \frac{2}{\pi} \frac{d}{ds} \int_s^\infty \frac{z \sin kz dz}{\sqrt{z^2 - s^2}} = -ks Y_0(ks), \quad (2.44)$$

$$\frac{d}{ds} \int_s^\infty \frac{z e^{-kz} dz}{\sqrt{z^2 - s^2}} = -ks K_0(ks) \quad (s > h) \quad (2.45)$$

and the k -derivative of the latter to obtain, after substitution of (2.28), (2.29), (2.41) and (2.42), a pair of integral equations. Then, seeking to mimic the disk and hole-in-a-wall calculations (Davis 1991, 1996), in which integrals from 0 to ∞ of $\cos kt \cos ks$, $\sin kt \sin ks$ yield multiples of $\delta(t - s)$, it is noted that $-2k^2(d/dk)(K_1 I_1)$ and $2(d/dk)(k^2 K_0 I_0)$ both $\rightarrow 1$ as $k \rightarrow \infty$ and

$$\int_0^\infty ks J_0(ks) J_0(kt) dk = \delta(t - s). \quad (2.46)$$

The corresponding result involving $ks Y_0(ks) Y_0(kt)$ must include the delta function because of the similar behaviour of the integrands as $k \rightarrow \infty$. Working from integrals given by (6.671) in Gradshteyn & Ryzhik (2007), it may be shown, via a lengthy

calculation, that

$$\int_0^\infty ksY_0(ks)Y_0(kt) dk = \delta(t-s) - \frac{4}{\pi^2} \frac{s}{s^2-t^2} \ln\left(\frac{s}{t}\right), \quad (2.47)$$

whence the transformed no-slip conditions (2.43) are identified as a pair of Fredholm integral equations of the second kind, namely

$$S(s) - \int_h^\infty [G_{11}(s, t)S(t) + G_{12}(s, t)T(t)] dt = F_1(s) = -\frac{2}{\pi s^{1/2}} \frac{d}{ds} \int_s^\infty \frac{U_1(1, z)z dz}{\sqrt{z^2 - s^2}}, \quad (2.48)$$

$$T(s) - \int_h^\infty [G_{21}(s, t)S(t) + G_{22}(s, t)T(t)] dt = F_2(s) = \frac{2}{\pi s^{1/2}} \frac{d}{ds} \int_s^\infty \frac{W_1(1, z)z dz}{\sqrt{z^2 - s^2}}, \quad (2.49)$$

in which the kernels are given by

$$G_{11}(s, t) = \sqrt{st} \int_0^\infty \left[1 + 2k^2 \frac{d}{dk} (K_1 I_1) \right] k J_0(ks) J_0(kt) dk \\ + \frac{4\sqrt{st}}{\pi} \int_0^\infty J_1^2 k^2 [2K_0(ks) - ksK_1(ks)][2K_0(kt) - ktK_1(kt)] dk, \quad (2.50)$$

$$G_{12}(s, t) = -2\sqrt{st} \int_0^\infty (K_0 I_0 - K_1 I_1) k^3 J_0(ks) Y_0(kt) dk \\ + \frac{4\sqrt{st}}{\pi} \int_0^\infty J_1 J_0 k^2 [2K_0(ks) - ksK_1(ks)][K_0(kt) - ktK_1(kt)] dk, \quad (2.51)$$

$$G_{21}(s, t) = G_{12}(t, s) \quad (2.52)$$

$$G_{22}(s, t) = \sqrt{st} \int_0^\infty \left[1 - 2 \frac{d}{dk} (k^2 K_0 I_0) \right] k Y_0(ks) Y_0(kt) dk + \frac{4}{\pi^2} \frac{\sqrt{st}}{s^2 - t^2} \ln\left(\frac{s}{t}\right) \\ + \frac{4\sqrt{st}}{\pi} \int_0^\infty J_0^2 k^2 [K_0(ks) - ksK_1(ks)][K_0(kt) - ktK_1(kt)] dk, \quad (2.53)$$

and the forcing functions are given by

$$F_1(s) = \frac{4s^{1/2}}{\pi} \int_0^\infty J_1 J_2 \{ -[K_0(ks) - ksK_1(ks)][e^{kh} + (3 + 2kh)e^{-kh}] \\ + K_0(ks)[(1 - kh)e^{kh} - (1 + kh)e^{-kh}] \} dk \quad (s > h), \quad (2.54)$$

$$F_2(s) = \frac{4s^{1/2}}{\pi} \int_0^\infty J_0 J_2 \{ [K_0(ks) - ksK_1(ks)][e^{kh} + (3 + 2kh)e^{-kh}] \\ - K_0(ks)[(2 - kh)e^{kh} + (2 + kh)e^{-kh}] \} dk \quad (s > h). \quad (2.55)$$

Evidently, the system is symmetric, that is, any unbiased discretization gives a linear system of equations whose matrix of coefficients is symmetric.

2.5. Wall pressure and stress

With S and T determined, the pressure and stress on the plane wall can be readily evaluated. The total wall pressure $P(r)$ is given, from (2.23) and (2.33), by

$$P(r) = P_1(r, 0) + P_2(r, 0) = 8\mu \int_0^\infty J_0(kr) J_2(k) e^{-kh} (2 + kh) dk$$

$$\begin{aligned}
& + 4\mu \int_h^\infty \int_0^\infty k^2 s^{1/2} J_0(kr) \{S(s)J_1(k)[ksK_1(ks) - K_0(ks)] \\
& - T(s)J_0(k)ksK_1(ks)\} dk ds,
\end{aligned} \tag{2.56}$$

and the wall stress $\tau(r)$ is, from (2.22) and (2.33),

$$\begin{aligned}
\tau(r) &= \mu \left[\frac{\partial U_1}{\partial z}(r, 0) + \frac{\partial U_2}{\partial z}(r, 0) \right] \\
&= -\frac{\mu}{r} L_{-1}(\Psi_1 + \Psi_2)(r, 0) = 8\mu \int_0^\infty J_1(kr)J_2(k)e^{-kh}(1 + kh) dk \\
&\quad + 4\mu \int_h^\infty \int_0^\infty k^2 s^{1/2} J_1(kr) \{S(s)J_1(k)[ksK_1(ks) - 2K_0(ks)] \\
&\quad - T(s)J_0(k)[ksK_1(ks) - K_0(ks)]\} dk ds.
\end{aligned} \tag{2.57}$$

2.6. Exit flow and wall contributions

It is of interest to identify the terms in (2.56) and (2.57) that arise from imposing no-slip at the plane wall. Suppose that Ψ_{rad}, Ψ_{ax} in (2.30) are replaced by ψ_{rad}, ψ_{ax} respectively, that is, velocities due to reflection at $z = 0$ are discarded. Then, without making the simplification involving image terms, the equations corresponding to (2.48) and (2.49) are

$$S(s) - \int_h^\infty [g_{11}(s, t)S(t) + g_{12}(s, t)T(t)] dt = f_1(s), \tag{2.58}$$

$$T(s) - \int_h^\infty [g_{21}(s, t)S(t) + g_{22}(s, t)T(t)] dt = f_2(s), \tag{2.59}$$

in which the kernels are given by

$$\begin{aligned}
g_{11}(s, t) &= \sqrt{st} \int_0^\infty \left[\frac{1}{2} + k^2 \frac{d}{dk} (K_1 I_1) \right] k [J_0(ks)J_0(kt) + Y_0(ks)Y_0(kt)] dk \\
&\quad + \frac{2}{\pi^2} \frac{\sqrt{st}}{s^2 - t^2} \ln \left(\frac{s}{t} \right),
\end{aligned} \tag{2.60}$$

$$g_{12}(s, t) = \sqrt{st} \int_0^\infty (K_0 I_0 - K_1 I_1) k^3 [Y_0(ks)J_0(kt) - J_0(ks)Y_0(kt)] dk, \tag{2.61}$$

$$g_{21}(s, t) = g_{12}(t, s), \tag{2.62}$$

$$\begin{aligned}
g_{22}(s, t) &= \sqrt{st} \int_0^\infty \left[\frac{1}{2} - \frac{d}{dk} (k^2 K_0 I_0) \right] k [J_0(ks)J_0(kt) + Y_0(ks)Y_0(kt)] dk \\
&\quad + \frac{2}{\pi^2} \frac{\sqrt{st}}{s^2 - t^2} \ln \left(\frac{s}{t} \right),
\end{aligned} \tag{2.63}$$

and the forcing functions are given by

$$f_1(s) = \frac{4s^{1/2}}{\pi} \int_0^\infty J_1 J_2 [ksK_1(ks) - khK_0(ks)] e^{kh} dk \quad (s > h), \tag{2.64}$$

$$f_2(s) = \frac{4s^{1/2}}{\pi} \int_0^\infty J_0 J_2 [(kh - 1)K_0(ks) - ksK_1(ks)] e^{kh} dk \quad (s > h). \tag{2.65}$$

This pair of integral equations determines the stream function for the exit flow adjustment. Comparison with (2.48) and (2.49) then identifies the terms introduced by the no-slip wall.

3. Numerical solution

For fixed h , S and T are determined from (2.48) and (2.49), which is a pair of Fredholm integral equation of the second kind and can be cast in a compact form: $\mathbf{c} - (\mathbf{K}\mathbf{c}) = \mathbf{F}$ where $\mathbf{c} = [S, T]^t$, $\mathbf{F} = [F_1, F_2]^t$ and $(\mathbf{K}\mathbf{c})(s) = \int_h^\infty \mathbf{G}(s, t)\mathbf{c}(t) dt$. Common methods for solving this type of equation are Nyström, Neumann and collocation (Atkinson 1997). But to ensure that the unacceptable homogeneous solution mentioned after (2.33) is suppressed, the Neumann method is adopted, in which the forcing terms are used to initiate an iterative scheme where $\mathbf{c} = \sum_{n=0}^\infty \mathbf{K}^n \mathbf{F}$ as long as $\|\mathbf{K}\| < 1$ (Atkinson 1997). Thus for J quadrature points and weights λ_i and w_i respectively, and with $\mathbf{c}_1(\lambda_i) = \mathbf{F}(\lambda_i)$ given, successive approximations to the discrete solution are determined for $n \geq 1$ by $\mathbf{c}_{n+1}(\lambda_i) = \sum_{j=1}^J w_j \mathbf{G}(\lambda_i, \lambda_j) \mathbf{c}_n(\lambda_j) + \mathbf{F}(\lambda_i)$.

The infinite integrals in $\mathbf{G}$ and $\mathbf{F}$ each involve the product of a well-behaved function involving modified Bessel functions and two Bessel functions of the first or second kind. These are calculated at the quadrature points within machine precision using IIPBF (Ratnanather *et al.* 2012) which is a MATLAB implementation of the Lucas algorithm (Lucas 1995), extended to also handle Bessel functions of the second kind. Appendix A.1 shows the asymptotic form of the kernels and forcing terms, whose short tails suggest that Gauss–Laguerre quadrature (Delves & Mohamed 1985) be used to compute λ_i and w_i via a MATLAB implementation (Burkardt 2010) of the IQPACK algorithm (Elhay & Kautsky 1987).

Collocation over $0 \leq r \leq 5$ and $5 \leq r \leq 15$ with 50 and 10 points respectively was used to compute (2.56) and (2.57). To accelerate the computations, the k -integrals associated with $S(s)$ in (2.56) and $T(s)$ in (2.57) are evaluated exactly as described in § A.2. For $r > 0$ IIPBF is used to compute the corresponding k -integral associated with $T(s)$ in (2.56) and $S(s)$ in (2.57), as well as the first integral in each equation. For $r = 0$, in (2.56) the first term is evaluated exactly as $8\mu \sin^3 \chi [(1 + \cos \chi)^{-2} + 1]$ where $h = \cot \chi$ (χ is the semi-angle subtended at the origin by the orifice), while the third term is computed using `quadgk`, which is MATLAB's adaptive quadrature function.

4. Results

Computations for $J = 64, 96, 112, 128$ yielded indistinguishable profiles for the wall pressure and wall shear stress. With the tolerance for the maximum absolute residual set as machine precision, the Neumann scheme converged in 10 iterations, thus confirming that $\|\mathbf{K}\| < 1$. Figures 2 and 3 respectively show, for $h = 1, 2, 3, 4, 5$ with $J = 64$, the dimensionless ratios of wall pressure and wall shear stress to viscosity, based on a length scale equal to the radius of the pipe, in which the mean velocity is 2 radii per unit time. Table 1 shows the computed values of $P(0)$, $\max \tau$ and $\arg \max \tau(r)$. The chosen values of h are of the same order as the pipe diameter, 2. Values outside this range are irrelevant to our envisioned physiological application.

5. Discussion

A motivating hypothesis based on observation of physiological experiments on isolated auditory sensory hair cells has been that an impinging jet on a plane wall results in a localized response in wall pressure and wall stress, especially if the inflow

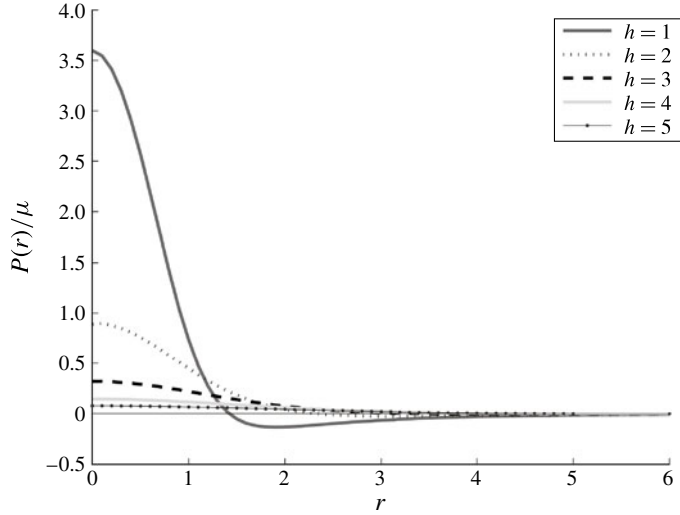


FIGURE 2. Dimensionless ratio of wall pressure, $P(r)$, to viscosity based on a length scale equal to the radius of the pipe in which the mean velocity is 2 radii per unit time for $h = 1, 2, 3, 4, 5$.

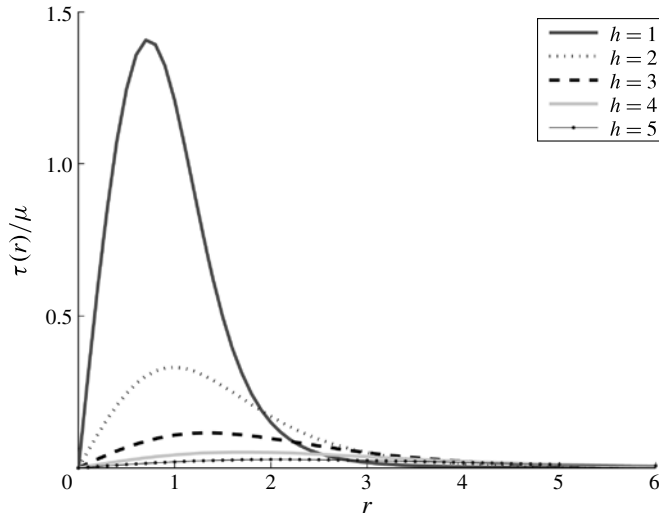
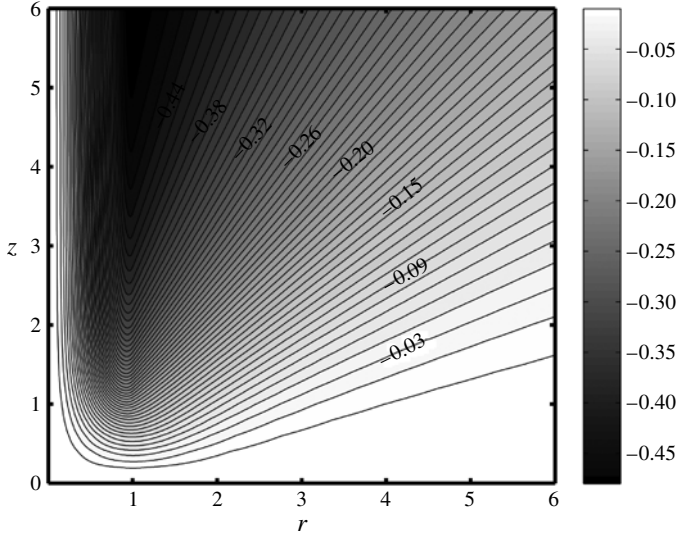


FIGURE 3. Dimensionless ratio of wall stress, $\tau(r)$, to viscosity based on a length scale equal to the radius of the pipe in which the mean velocity is 2 radii per unit time for $h = 1, 2, 3, 4, 5$.

is close to the wall. Such a response was first modelled for physiological flows by a numerical solution of Navier–Stokes equations (Deshpande & Vaishnav 1982, 1983) for a range of Reynolds numbers from 1 to 2000. Their inclusion of results for unity Reynolds number is questionable, as their method assumes the presence of convection and a parabolic exit flow that renders unnecessary the consideration of flow within the pipe. That is inapplicable here, because the known presence of the Papkovitch–Fadle

h	1	2	3	4	5
$P(0)$	3.5975	0.8981	0.3180	0.1433	0.0757
$\max \tau$	1.4069	0.3310	0.1147	0.0514	0.0271
$\arg \max \tau(r)$	0.7	1.0	1.3	1.7	2.1

TABLE 1. $P(0)$, $\max \tau$ and $\arg \max \tau(r)$ for $h = 1, 2, 3, 4, 5$.FIGURE 4. Contour representation of Ψ_1 for $h = 1$ from (2.20) computed via the same strategy as for wall pressure and wall shear stress. Note the seepage through the pipe wall. Contour shading is graded in the negative direction from zero at $z = 0$.

pipe eigenfunctions is evidence that the pipe flow exhibits significant adjustments as it approaches the orifice. Though it is not of physiological interest, the entire flow field i.e. $\Psi_1 + \Psi_2$ can be computed via the same strategy for the wall pressure and wall shear stress. Indeed, figure 4 illustrates the structure of Ψ_1 in the neighbourhood of the orifice for $h = 1$. Clearly, in the interior near $z = 1$ the streamlines hit the pipe wall but are almost parallel at $z = 5$. The far-field structure described by (2.27) is recovered, which is independent verification of the IIPBF toolbox used in the computational strategy. The role of Ψ_2 (not shown) is to eliminate the pipe wall seepage generated by Ψ_1 . Its magnitude is such that it modifies the flow field near the orifice, and thus it suffices to conclude that the stream function structure derived here is valid throughout the pipe interior and does not induce the eddy structure at the level of the orifice observed by Deshpande & Vaishnav (1982) when convection is present.

The computations for the wall pressure and wall stress presented here are in accordance with the physics of the flow. First, the plane wall pressure is always at a maximum at $r = 0$, which decreases as h increases to alleviate the obstruction effect of the wall. Second, owing to the interaction of the inflow with the ambient fluid in the neighbourhood of $z = 0$, the wall stress rises rapidly to a maximum and then

decays; the radial position of that maximum increases as h increases. This behaviour is also observed in non-zero Reynolds number flows; indeed the characteristics of axisymmetric stagnation point flow (e.g. Phares *et al.* 2000b) can be obtained from (2.56) and (2.57) for $r \ll 1$.

The analytical approach, assuming that the flow is highly viscous, is based on a Stokes-flow model and uses suitably modified Stokeslet singularities distributed along the pipe wall. A lengthy reformulation of the no-slip requirement into a pair of Fredholm integral equations permits the derivation of expressions for the induced wall pressure and wall stress.

With respect to physiological experiments on auditory sensory hair cells, specifically perfusing cochlear outer hair cells (OHCs) via a pipette, poration (i.e. pore formation) is a likely but unproven scenario (Brownell 2006). Supporting evidence comes from the observation that it took less than 3 minutes for a fluorescent dye to enter the OHC when perfused with a pipette compared with 120 minutes via a flow chamber, which is akin to application via a pipette at a large distance (Zhi, Ratnanather & Brownell 1995). In equilibrium, pores open and close in a random manner. Poration occurs when the intrinsic pore line tension is matched by the uniform resultant wall stress, as demonstrated by models of electrically induced poration or electroporation (Chen *et al.* 2006). A model of mechanically induced poration could incorporate the non-uniform resultant stress derived above into the Saffman equation for a viscous fluid sheet (Saffman 1976) in a manner similar to Fujitani (2009). Hence the closer the orifice is to the wall, the greater likelihood of mechanoporation resulting in the observed stronger cellular response to the perfusate. This concept of mechanoporation and its application in the wider area of cellular physiology needs to be explored more fully. Finally, it will be interesting to see how the wall stress and wall pressure are affected by mass transfer (Nakagawa *et al.* 1990; Housley *et al.* 1992; Lin *et al.* 1993; Santos-Sacchi *et al.* 2006), permeable plane wall (Ratnanather *et al.* 1996; Morimoto *et al.* 2002; Zhi *et al.* 2007), slip on the plane wall (Lauga, Brenner & Stone 2007) and oscillating inflow (Brundin & Russell 1994).

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Appendix. Asymptotic properties and exact integrals

A.1. Asymptotic properties of kernels and forcing terms

Asymptotic properties of the kernels and forcing terms were used to verify the need for Gauss–Laguerre quadrature to solve the Fredholm integral equation (Delves & Mohamed 1985).

A.1.1. G_{11}

$$G_{11}(s, t) = G_{11a}(s, t) + G_{11b}(s, t), \quad (\text{A } 1)$$

where

$$G_{11a}(s, t) = \sqrt{st} \int_0^\infty \left[1 + 2k^2 \frac{d}{dk} (K_1 I_1) \right] k J_0(ks) J_0(kt) dk, \quad (\text{A } 2)$$

$$G_{11b}(s, t) = \frac{4\sqrt{st}}{\pi} \int_0^\infty J_1^2 k^2 [2K_0(ks) - ksK_1(ks)][2K_0(kt) - ktK_1(kt)] dk. \quad (\text{A } 3)$$

But

$$\begin{aligned} & \sqrt{st} \int_L^\infty \left[1 + 2k^2 \frac{d}{dk} (K_1 I_1) \right] k J_0(ks) J_0(kt) dk \\ & \sim \sqrt{st} \int_L^\infty \frac{9}{8k} \frac{\pi}{2k\sqrt{st}} \cos(ks - \pi/4) \cos(kt - \pi/4) dk \\ & = \frac{9\pi}{32} \int_L^\infty \frac{1}{k^2} [\sin k(s+t) + \cos k(s-t)] dk, \\ & = \frac{9\pi}{32} \left[\frac{\cos L(s+t)}{L^2(s+t)} + O\left(\frac{1}{L^3}\right) + \frac{\cos L(s-t)}{L} + (s-t) \text{Si}[L(s-t)] \right], \quad (\text{A } 4) \end{aligned}$$

which provides an estimate of the tail of the integral. Here Si denotes the sine integral, defined by

$$\text{Si}(u) = \int_u^\infty \frac{\sin x}{x} dx. \quad (\text{A } 5)$$

For $s = t$, use can be made of the integral 6.535 in Gradshteyn & Ryzhik (2007) to write

$$\begin{aligned} G_{11a}(s, s) &= s \int_0^\infty \left[1 + 2k^2 \frac{d}{dk} (K_1 I_1) - \frac{9}{9 + 8k^2} \right] k J_0(ks) J_0(kt) dk \\ &\quad + \frac{9s}{8} I_0\left(\frac{3s}{2\sqrt{2}}\right) K_0\left(\frac{3s}{2\sqrt{2}}\right), \quad (\text{A } 6) \end{aligned}$$

in which the tail is $O(L^{-3})$.

The modified Bessel functions in G_{11b} ensure that the integrand decays exponentially.

A.1.2. G_{12}

$$G_{12}(s, t) = G_{12a}(s, t) + G_{12b}(s, t) + G_{12c}(s, t), \quad (\text{A } 7)$$

where

$$G_{12a}(s, t) = -2\sqrt{st} \int_0^\infty \left[(K_0 I_0 - K_1 I_1) k^3 - \frac{1}{4} \right] J_0(ks) Y_0(kt) dk, \quad (\text{A } 8)$$

$$G_{12b}(s, t) = \frac{4\sqrt{st}}{\pi} \int_0^\infty J_1 J_0 k^2 [2K_0(ks) - ksK_1(ks)][K_0(kt) - ktK_1(kt)] dk, \quad (\text{A } 9)$$

$$G_{12c}(s, t) = -\frac{\sqrt{st}}{2} \int_0^\infty J_0(ks) Y_0(kt) dk. \quad (\text{A } 10)$$

The first and third terms exploit the large k behaviour $(K_0 I_0 - K_1 I_1) k^3 \sim (1/4) + (9/32k^2)$, by using the evaluation

$$G_{12c}(s, t) = 0 \quad (s < t), \quad (\text{A } 11)$$

$$G_{12c}(s, s) = 1/4, \quad (\text{A } 12)$$

$$\begin{aligned}
G_{12c}(s, t) &= \frac{1}{\pi} \int_0^1 \sqrt{\left(\frac{1 - \beta^2}{1 - \beta^2 z^2} \right)} \frac{dz}{\sqrt{1 - z^2}} \\
&= \frac{\sqrt{1 - \beta^2}}{\pi} F(\pi/2, \beta) \quad (s > t, \beta^2 = 1 - t^2/s^2). \tag{A 13}
\end{aligned}$$

The value at $s = t$ is the average at the discontinuity and immaterial in the integral equations. The elliptic integral of the first kind is defined by

$$F(\pi/2, \beta) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - \beta^2 \sin^2 \phi}}. \tag{A 14}$$

The ‘tail’ of the G_{12a} integral is

$$\begin{aligned}
&-2\sqrt{st} \int_L^\infty \left[(K_0 I_0 - K_1 I_1) k^3 - \frac{1}{4} \right] J_0(ks) Y_0(kt) dk \\
&\sim \frac{9\pi}{64} \int_L^\infty \frac{1}{k^3} [\cos k(s+t) - \sin k(s-t)] dk = O(L^{-3}). \tag{A 15}
\end{aligned}$$

The integrand in G_{12b} decays exponentially.

A.1.3. G_{21}

Note that $G_{21}(s, t)$ is obtained from $G_{12}(s, t)$ by interchange of s, t . If the symmetry is exploited by working with $s \geq t$ then, in effect, only the J_0, Y_0 in the first and third integrals need be interchanged.

A.1.4. G_{22}

$$G_{22}(s, t) = G_{22a}(s, t) + G_{22b}(s, t) + G_{22c}(s, t), \tag{A 16}$$

where

$$G_{22a}(s, t) = \sqrt{st} \int_0^\infty \left[1 - 2 \frac{d}{dk} (k^2 K_0 I_0) \right] k Y_0(ks) Y_0(kt) dk, \tag{A 17}$$

$$G_{22b}(s, t) = \frac{4\sqrt{st}}{\pi} \int_0^\infty J_0^2 k^2 [K_0(ks) - ks K_1(ks)] [K_0(kt) - ks K_1(kt)] dk, \tag{A 18}$$

$$G_{22c}(s, t) = \frac{4}{\pi^2} \frac{\sqrt{st}}{s^2 - t^2} \ln \left(\frac{s}{t} \right). \tag{A 19}$$

The ‘tail’ of the G_{22a} integral is

$$\begin{aligned}
&\sqrt{st} \int_L^\infty \left[1 - 2 \frac{d}{dk} (k^2 K_0 I_0) \right] k Y_0(ks) Y_0(kt) dk \\
&\sim \frac{\pi}{32} \int_L^\infty \frac{1}{k^2} [\cos k(s-t) - \sin k(s+t)] dk, \tag{A 20}
\end{aligned}$$

which differs from the ‘tail’ of the G_{11a} integral by $O(L^{-2})$.

The integrand in G_{22b} decays exponentially.

A.1.5. F_1 and F_2

On inserting the $k \ll 1$ approximations for the functions of k ,

$$F_1(s) = \frac{4s^{1/2}}{\pi} \int_0^\infty J_1 J_2 \{ -[K_0(ks) - ks K_1(ks)] 2[\cosh kh + (1 + kh)e^{-kh}] \}$$

$$\begin{aligned}
& + K_0(ks)2(\sinh kh - kh \cosh kh)\} dk \\
& \sim -\frac{s^{1/2}}{\pi} \int_0^\infty k^3 [K_0(ks) - ksK_1(ks)] dk = \frac{12}{\pi s^{7/2}}, \quad (A 21)
\end{aligned}$$

by use of

$$\int_0^\infty x^{n+2} K_{n-1}(x) dx = 2 \int_0^\infty x^{n+1} K_n(x) dx = \lim_{x \rightarrow 0} [x^{n+1} K_{n+1}(x)] = n! 2^{n+1}. \quad (A 22)$$

Similarly,

$$\begin{aligned}
F_2(s) &= \frac{4s^{1/2}}{\pi} \int_0^\infty J_0 J_2 \{ [K_0(ks) - ksK_1(ks)] 2[\cosh kh + (1 + kh)e^{-kh}] \\
&\quad - K_0(ks) 2(2 \cosh kh - kh \sinh kh) \} dk \\
&\sim -\frac{2s^{1/2}}{\pi} \int_0^\infty k^3 s K_1(ks) dk = -\frac{3}{s^{5/2}}, \quad (A 23)
\end{aligned}$$

by use of

$$\int_0^\infty x^\mu K_\nu(ax) dx = \left(\frac{2}{a}\right)^\mu \frac{1}{2a} \Gamma\left(\frac{1+\mu+\nu}{2}\right) \Gamma\left(\frac{1+\mu-\nu}{2}\right) \quad (A 24)$$

(see 6.561/16 in Gradshteyn & Ryzhik 2007). The μ -derivative of this is also needed to estimate G_{11} for $t = O(1)$. The first integral is exponentially small and the second of order $s^{-9/2} \ln s$.

A.2. Exact integrals for wall pressure and wall stress

The k -integrals associated with $S(s)$ in (2.56) and $T(s)$ in (2.57) are evaluated as follows. According to (6.525) in Gradshteyn & Ryzhik (2007),

$$\begin{aligned}
\int_0^\infty k^2 J_1(kr) K_0(ks) J_0(kr') dk &= \frac{2r(r^2 + s^2 - r'^2)}{[(r^2 + s^2 + r'^2)^2 - 4r^2 r'^2]^{3/2}} \\
&= \frac{1}{[(r^2 + s^2 + r'^2)^2 - 4r^2 r'^2]^{1/2}} \left[\frac{r - r'}{(r - r')^2 + s^2} + \frac{r + r'}{(r + r')^2 + s^2} \right]. \quad (A 25)
\end{aligned}$$

Hence, by differentiation with respect to s ,

$$\begin{aligned}
& \int_0^\infty k^3 s J_1(kr) K_1(ks) J_0(kr') dk \\
&= \frac{4rs^2}{[(r^2 + s^2 + r'^2)^2 - 4r^2 r'^2]^{3/2}} \left\{ \frac{3[(r^2 + s^2)^2 - r'^4]}{(r^2 + s^2 + r'^2)^2 - 4r^2 r'^2} - 1 \right\} \\
&= \frac{s^2}{[(r^2 + s^2 + r'^2)^2 - 4r^2 r'^2]^{1/2}} \left\{ \frac{2(r - r')}{[(r - r')^2 + s^2]^2} + \frac{2(r + r')}{[(r + r')^2 + s^2]^2} \right. \\
&\quad \left. + \left[\frac{1}{(r - r')^2 + s^2} + \frac{1}{(r + r')^2 + s^2} \right] \left[\frac{r - r'}{(r - r')^2 + s^2} + \frac{r + r'}{(r + r')^2 + s^2} \right] \right\}. \quad (A 26)
\end{aligned}$$

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